



PROFESSIONAL SUMMARY

Computer Science student at RMIT University, minoring in Artificial Intelligence & Machine Learning, who ships production software end to end. Built and sold a medical practice management platform now in daily clinical use, plus four further full-stack products across web, mobile and Python — several with real users, all integrating live third-party and AI APIs. Seeking a software engineering internship or graduate role.

PROJECTS

Medical Practice Management Platform

June 2026 – Present

Built independently; now licensed to the practice on a paid monthly subscription

React · TypeScript · Supabase (PostgreSQL) · Deno · Capacitor · Tailwind CSS — private repository, walkthrough available on request

- Built and sold a full-stack platform now in daily clinical use at a Jakarta family practice, turning a self-initiated project into recurring monthly revenue; it automates the manual record-keeping I previously did by hand at that same practice.
- Grown to 66 serverless Deno edge functions and 31 PostgreSQL migrations across a TypeScript monorepo spanning a React dashboard and two Android apps (Capacitor), covered by 269 unit tests.
- Built an insurance claims module that fills insurers' own reimbursement PDFs — by field name for AcroForms, by a percentage-based coordinate map for scanned ones — signed and stamped only once the doctor approves.
- Integrated the WhatsApp Business API for bilingual (English / Bahasa Indonesia) invoices, prescriptions, reminders, appointment booking and claim intake, plus Google Calendar and Drive for scheduling and per-patient document storage.
- Added an in-app notification bell plus Android push via Firebase Cloud Messaging, configurable per channel.
- Integrated Claude (Opus 5) as an eight-tool assistant that books appointments and drafts documents, and to draft consultation reports and medical certificates — all behind a propose-then-approve flow, so nothing clinical is sent without the doctor's review.

Locked In — Social Fitness App

April – May 2026

React Native (Expo) · TypeScript · Supabase · Gemini API · EAS Build · Vercel — github.com/TusharLachman25/workout-tracker

- Designed and shipped a squad-based workout tracker in daily use by my friend group, distributed as an Android APK through EAS Build with a companion web build auto-deploying to Vercel.
- Built stories, leaderboards, a social feed, direct messaging and push notifications on Supabase using Auth, Storage, Realtime subscriptions and row-level security policies.
- Integrated Gemini vision to read workout screenshots from smartwatch apps and auto-fill activity, distance, duration, pace and calories, removing manual data entry.
- Documented environment setup, release builds and schema-migration practice so others could contribute.

Kitchen OS — Pantry & Nutrition Assistant

February – March 2026

Python · Streamlit · Supabase (PostgreSQL) · Google Gemini · pandas · Altair — github.com/TusharLachman25/AI-Meal-Suggestion

- Built a pantry, recipe and macronutrient tracking app in ~3,500 lines of Python, on a relational schema designed from an entity-relationship diagram.
- Implemented automatic pantry deduction when a recipe is cooked, including unit normalisation and conversion between weight, volume and count.
- Integrated Gemini for a conversational recipe assistant with persistent multi-session chat history, and vision-based food-photo logging that estimates macros from an image.

Other Projects

2026

- **Jarvis — Personal Life OS** (private repo) · Next.js, Supabase, Claude API — a dashboard unifying tasks, calendar, finance and health over 38 REST endpoints, reconciling two wearables whose health platforms don't interoperate.
- **Bets** (private repo) · React Native, Supabase — a mobile app for logging and settling informal bets within a friend group.

Python · Flask · boto3 · Docker · AWS · NetworkX · transformers · Clingo (ASP) · Prolog

- **Cloud:** built a Flask REST API on DynamoDB and S3 and deployed it two ways for comparison — directly on EC2 with provisioning scripts, and containerised with Docker and gunicorn on ECS.
- **Social media analytics (team):** a six-stage pipeline over YouTube API data — VADER and RoBERTa sentiment, LDA and BERTopic topic modelling, then NetworkX centrality, Louvain communities and homophily.
- **Constraint optimisation:** exam timetabling solved in Answer Set Programming with Clingo against a benchmark suite; ML coursework in scikit-learn on regression and multi-class classification.

EXPERIENCE

Information Systems Intern

August 2023 – January 2024

Makmur Jaya Abadi | Jakarta, Indonesia

- Digitized and organized accounting records into Excel and Google Sheets, ensuring data accuracy and accessibility for analysis.
- Developed and implemented automated solutions using formulas and advanced spreadsheet features, resulting in a reduction in manual data entry time.
- Collaborated with accounting staff to identify areas for process improvement and implement automation strategies.

Part Time Medical Assistant

June 2020 – April 2023

Dr Varkha's Teleconsultations | Jakarta, Indonesia

- Digitized patient medical records.
- Compiled laboratory and x-ray reports, patient notes and other pertinent data.
- Updated patient medical records in system, carefully protecting patient information and verifying data.
- Supported medical staff by answering phones and making appointments.

EDUCATION

Bachelor of Computer Science — Minor in Artificial Intelligence & Machine Learning

Expected July 2027

RMIT University, Melbourne, Australia

Elementary School, Middle School and High School

Completed April 2023

National High Jakarta School, Jakarta, Indonesia

- Completed Cambridge IGCSE, AS Level and A Level examinations.

Balvikas Student

August 2011 – August 2023

Sathya Sai Education, Jakarta, Indonesia

SKILLS

- **Languages:** Python, TypeScript, JavaScript, Java, C, C++, SQL, Prolog, HTML, CSS
- **Frameworks:** React, React Native (Expo), Next.js, Node.js, Flask, Streamlit, Tailwind CSS, Capacitor
- **Cloud & DevOps:** AWS (EC2, ECS, S3, DynamoDB), Docker, Vercel, EAS Build, Firebase Cloud Messaging, Git/GitHub
- **Backend & Data:** PostgreSQL, Supabase, REST API design, serverless functions, OAuth 2.0, schema design and migrations
- **AI & Data Science:** Claude API, Google Gemini (text and vision), scikit-learn, pandas, NLTK, transformers, NetworkX

COMMUNITY SERVICE & INTERESTS

The Care Club — volunteer every 2-4 weeks with a charity group providing food, groceries, haircuts and medical services to families in need, alongside a general practitioner, optician and dentist.

LANGUAGE

English

Proficient (C2)

Bahasa Indonesia

Advanced (C1)

Hindi

Beginner (A1)

Mandarin

Beginner (A1)